

sorteren + opmaken

Omschrijving	sorteren + opmaken		Component	Opmerking	Waarde	Vorm	Distributeur	#	€ (1)	#	€€	Symbool	Footprint	link
	Type													
FRAM chip	IC, SMD	FM24V01A-GTR	128-Kbit (16Kbyte)	-	8-SOIC	Digikey	1	€ 3,82	3,82	https://www.mouser.be/ProductDetail/Microchip-Technology/24LC02-100-ND	https://www.mouser.be/ProductDetail/Microchip-Technology/24LC02-100-ND	https://www.digikey.be/nl/products/detail/infinity		
STM32	Knop, THT	TS02-66-43-BK-160-LCR-D		-	SPST momentary tactile, 6 pins	Mouser	2	€ 0,09	0,172	Switch:SW_Push	s	https://www.mouser.be/ProductDetail/SameStar/TS02-66-43-BK-160-LCR-D-ND		
STM32	L, SMD	BLM18KG601SN1D	600 Ω @ 100 MHz; filtert VDDA-pin M	600Ω@100MHz	0603	Digikey	1	€ 0,09	0,09	Device:Ferrite_Bead_Small	Inductor_SMD:L_0603_1608Metric Inductor_SMD:L_0603_1608Metric	https://www.digikey.be/en/products/detail/murphy		
BMS - li-ion lader, BMS - 3,3V buck-boost	L, SMD	MAKK2520H1R0M	Datasheet: 1 µH typ, I_sat ≥ 2A	1µH	Shielded power, 2,5x2,0mm	Mouser	3	€ 0,16	0,489	Device:L	Inductor_SMD:L_Taiyo-Yuden_NR-2 Inductor_SMD:L_Taiyo-Yuden_NR-2 https://www.mouser.be/ProductDetail	https://www.mouser.be/ProductDetail/Taiyo-Yuden		
LED	LED	BL-BEG204-7-E	Rood en groen, difuus, THT, common	-	5mm LED, radiaal, 3 pinnen	Digikey	1	€ 0,17	0,17	LED_DUAL_KAK	LED_D5.0mm-3_Horizontal_O3.81mm	https://www.digikey.be/nl/products/detail/ame		
GPS/GNSS	Module	Quectel LC76GPAMD	Zuiniger, goedkoper bij groot aantal		half hole PCB breakout (1.1mm)	Mouser	1	€ 7,76	7,76	https://www.mouser.be/ProductDetail/Quectel/	https://www.mouser.be/ProductDetail/Quectel/	https://www.mouser.be/ProductDetail/Quectel/		
Accelerometer	Module	LIS2DW12 module	module bevat I²C pullups weerstande	-	Module (2 4-pin headers)	Tinytronics	1		0	Zelf gemaakt	! zelf maken	https://www.tinytronics.nl/en/sensors/accelera		
SD-kaart	Module	SD adapter module	Module met level shifter en SD-houder	-	Module (8x2 pin header)	Tinytronics	1	€ 2,50	2,5	Zelf gmaakt	PinHeader_2x08_P2.54mm_Horizontal PinHeader_2x08_P2.54mm_Vertical	https://www.tinytronics.nl/en/data-storage/mo		
E-ink	Module, Extern	Waveshare 1.54" module	-		8-pin JST PH 2.0mm aansluiting	Waveshare	1		0	Zelf gemaakt	zie connector	https://www.waveshare.com/1.54inch-e-Paper-		
ST-link	Module, Extern, alternatief	STLINK-V3MINIE	Programmer/debugger	-	Standard Card Edge Connector	Farnell	1	€ 28,47	28,4713	https://www.mouser.be/ProductDetail/STMicroelectronics/STLINK-V3MINIE-ND	Zie connector	https://be.farnell.com/stmicroelectronics/stlink		
LoRa	Module, SMD	WisDuo RAK11720 (MHF4)	LoRa + BLE, MHF4 connector		half hole PCB breakout (1.1mm)	Digikey	1	€ 12,76	12,76	https://www.snapeda.com/parts/R	https://www.snapeda.com/parts/R	https://www.digikey.be/nl/products/detail/rakw		
Motor	Motor, Extern	TD-8120MG Digital Servo	waterproof - 20kg	-	Zie datasheet	Tinytronics	1	€ 10,25	10,25	Zelf gemaakt	Zie connector	https://www.tinytronics.nl/en/mechanics-and-a		
BMS - li-ion lader	NTC, THT, alternatief	103AT-2	THT, duurder, makkelijker tegen batterij	10K @25°C,	Radial, 2,54mm spacing	Mouser	1	€ 0,58	0,583	Device:Thermistor_NTC	Capacitor_THT:C_Disc_D3.4mm_W	https://www.mouser.be/ProductDetail/Semitec		
STM32	R, SMD	RC0603FR-07220KL	VBAT_SENSE-deler: 220K/(220K+100K)	220K	0603	Mouser	1	€ 0,09	0,086	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric	https://www.mouser.be/ProductDetail/YAGEO/		
STM32, BMS - 5V boost	R, SMD	RC0603FR-07100KL	Zie hierboven	100K	0603	Mouser	3	€ 0,09	0,258	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric	https://www.mouser.be/ProductDetail/YAGEO/		
GPS/GNSS	R, SMD	RC0603FR-131KL	Om supercap op te laden	1K	0603	Mouser	1	€ 0,09	0,086	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric	https://www.mouser.be/ProductDetail/YAGEO/		
GPS antenne	R, SMD	RC0402JR-070RL	zie hardware design p.40	0R	0402	Mouser	1	€ 0,09	0,086	Device:R	Resistor_SMD:R_0402_1005Metric Resistor_SMD:R_0402_1005Metric	https://www.mouser.be/ProductDetail/YAGEO/		
BMS - li-ion lader	R, SMD	RC0603FR-072K2L		2,2K	0603	Mouser	1	€ 0,09	0,086	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric	https://www.mouser.be/ProductDetail/YAGEO/		
BMS - li-ion lader	R, SMD	RC0603FR-07100RL		100R	0603	Mouser	1	€ 0,09	0,086	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric	https://www.mouser.be/ProductDetail/YAGEO/		
BMS - li-ion lader	R, SMD	RC0603FR-075K23L	JEITA-pullup met 103AT NTC (zie BQ	5,23K	0603	Mouser	1	€ 0,09	0,086	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric	https://www.mouser.be/ProductDetail/YAGEO/		
BMS - 3,3V buck-boost	R, SMD	RC0603FR-071ML		1M	0603	Mouser	1	€ 0,09	0,086	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric	https://www.mouser.be/ProductDetail/YAGEO/		
BMS - 3,3V buck-boost	R, SMD	RC0603FR-07294KL	1% tolerantie aangeraden voor Vout-p	294K	0603	Mouser	1	€ 0,09	0,086	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric	https://www.mouser.be/ProductDetail/YAGEO/		
Motor	R, SMD	PT1206FR-7W0R1L	Stall-stroom servo ≈ 1A; bevestiging wa	0,1R / 500mA	1206	Mouser	1	€ 0,32	0,318	Device:R	Resistor_SMD:R_1206_3216Metric Resistor_SMD:R_1206_3216Metric	https://www.mouser.be/ProductDetail/YAGEO/		
Motor	R, SMD	RC0603FR-071KL	RC filter met 4,8kHz cutoff	1K	0603	Mouser	1	€ 0,09	0,086	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric	https://www.mouser.be/ProductDetail/YAGEO/		
LED	R, SMD	AF0603FR-07160RL		160R	0603	Mouser	1	€ 0,09	0,086	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric	https://www.mouser.be/ProductDetail/YAGEO/		
LED	R, SMD	RC0603FR-07130RL		130R	0603	Mouser	1	€ 0,10	0,095	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric	https://www.mouser.be/ProductDetail/YAGEO/		
STM32, LoRa, GPS/GNSS, SD-kaart, E	R, SMD, alternatief	RC0603FR-1310KL	-	10K	0603	Mouser	9	€ 0,09	0,774	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric	https://www.mouser.be/ProductDetail/YAGEO/		
BMS - li-ion lader	R, SMD, alternatief	RC0603FR-1330K1L	Parallel met NTC voor JEITA-temperatu	30,1K	0603	Mouser	1	€ 0,09	0,086	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric	https://www.mouser.be/ProductDetail/YAGEO/		
GPS/GNSS	TVS, SMD	SMF05CT1G	5 TVS dioden in één, 2 gebruikt	-	SC-88-6 (SOT-363-6)	Mouser	1	€ 0,38	0,378	https://www.snapeda.com/parts/S	https://www.snapeda.com/parts/S	https://www.mouser.be/ProductDetail/onsemi/		
GPS antenne	TVS, SMD	ESD9B5.0ST5G	Lage junction-capacitance (<0,6pF) — zie hardware		SOD-923	Digikey	1	€ 0,19	0,19	Device:D_TVS	Diode_SMD:D_SOD-923	https://www.digikey.be/en/products/detail/ons		
BMS - USB-C	TVS, SMD	USBLC6-2SC6	Plaatsen direct naast USB-C connect	-	SOT-23-6	Digikey	1	€ 0,16	0,16	Device:D_TVS	Package_TO_SOT_SMD:SOT-23-6 Package_TO_SOT_SMD:SOT-23-6	www.digikey.be/nl/products/detail/stmicroelec		
Totaal							0	€ 46.261,51	120,4573					

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	Type												